Using Machine Learning and Computer Vision to Analyze and Improve Running Form

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Engineering Problem and Goal

Running is an affordable, accessible, and efficient cardiovascular exercise for individuals looking to improve their physical and mental health. While it is highly accessible, injuries sustained due to improper form can dissuade runners from continuing to exercise, especially at an amateur level.

- Around the world and especially in the United States, individuals suffer from degrading physical health due to a lack of cardiovascular exercise. Contemporary research has linked the global mental health epidemic to this degrading physical health (Ratey & Hagerman 4). An increase in cardiovascular exercise would largely improve physical and mental health, as cardiovascular exercise has been shown to be a powerful antidepressant and mood stabilizer (Ratey & Hagerman 114-118).

- Running is ideal cardiovascular exercise, as it is accessible to most people. The barriers to entry for gear and location are very low - all a person needs to begin running is a place to run outside and a pair of sneakers. However, running injuries are incredibly common, especially to new runners who have not developed proper form (Zhang, Guo, & Zanotto). Removing these barriers to entry would allow more people to begin running, which would in turn improve physical and mental health for those who abandon running due to injury.

Engineering Problem and Goal

- There is a large body of contemporary research that deals with machine learning and gait analysis. Some of it is focused specifically on running or improving gait in general, but there is also research being done on using ML to rehabilitate individuals with physical or neurological conditions that negatively impact their gait.

- This ML-gait analysis confluence mostly trains using data derived from sensors, as opposed to computer vision data. Combining computer vision and machine learning together to analyze gait is a more novel approach from a research perspective (Khera & Khumar). From a product perspective, there exist numerous computer vision-based programs which analyze and offer critique on running gait, such as GaitOn and Running3DMA. However, these do not use machine learning.

- The two halves of this problem - the research half, and the product half - each use roughly half of the technology necessary for the goal program. Thus the main objective for this project is combining those two halves together, in order to create a machine learning computer vision hybrid program which can analyze and improve a runner's gait.

Procedures

- The primary method for constructing this model is continual testing and prototyping of an ML-CV hybrid model, which will be iteratively built using a dataset of running videos found online. The primary components for this project are the dataset, OpenCV, and Google MediaPipe. The dataset being used is the running videos found in the Kinetics dataset from DeepMind, which is a larger library containing hundreds of thousands of humans completing various motion-based tasks. Only the running subset of these videos is being used, which totals to roughly 800 10-second videos.
- The initial step was creating some sort of program which would be able to pull motion-based data from a video of a person running. This program was created in Python using OpenCV, and it allows any video to be turned in to numerical data representing each of the joints in a runner's body throughout the course of their run. This extraction of data allows the video, containing visual data, into a form which can then be fed into a support vector machine. The next step is to tag each video in the Kinetics dataset as exhibiting proper or improper form before feeding those videos into the extraction program.
- After each video has had motion-based data extracted, the motion-based data is given to the SVM along with the tag of "proper/improper" form. From there, the machine can adequately judge whether or not a runner is running correctly or incorrectly. While an unsupervised machine learning model was considered as it would bypass the need for manual tagging by the researcher, eventually a supervised machine learning model was selected because of the researcher's familiarity with supervised ML and the time constraints that the project featured. The motion-based data means that the user can upload a video of themselves, but the ML model does not require any convolutional layers since all of the data is numerical.
- Keras, the ML design library which was used in the course of this project, provides built-in features to measure the accuracy of a neural network. Because this project uses supervised machine learning where a correct "answer key" is given to the network, the network can give an accuracy score to itself. The initial objective was to achieve a testing accuracy of over 85%. Although that objective was not met initially, various changes can be made to the model such as processing the videos in such a way that the runner is more easily detected by OpenCV and MediaPipe, splitting up the videos of runners into shorter or longer segments, or standardizing the data around some central mean to account for the different position of the runner in the discrete videos.

Results

- The initial result shows a 67% test accuracy for by running the dataset with 825 data points, each with 2277 features, through a linear SVM (Support Vector Machine). While this is below the initial engineering goal, the discussion includes several sources of error for how this figure can be improved.

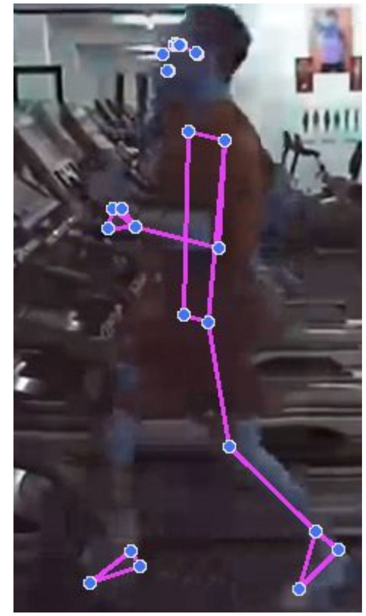
```
             -------------  --------------
             Predicted Bad  Predicted Good
Actual Bad   243            130
Actual Good  72             380
-----------  -------------  --------------
-----------  -----
Accuracy:    0.755
Error Rate:  0.245
Precision:   0.745
Recall:      0.841
```

- This figure shows a confusion matrix for the model, run on the entire dataset of classified videos
- The initial research goal of 85% test accuracy was not achieved, but the results shown, as well as other aspects of the project, demonstrate the possible room for improvement moving forward that could be achieved

Discussion

- The key interpretation of these initial results is that they demonstrate the feasibility of this novel approach to gait analysis. Most other studies incorporate some form of kinematic sensor when working with gait analysis, as that type of data is usually better suited to a machine learning project (Simone et. al, 2020) (Lau et. al, 2008). This project uses raw visual input, which is then transformed into numerical data.
- This allows the user, if such an application is eventually pursued for this research, to test their own form without having access to a kinematic sensor. This pushes the accessibility of this technology forward considerably, and opens the way for individuals to analyze their running form using their own smartphone, for example.
- One key thing to note about these results is that there is a relatively small sample size, which is affected by the programs that were used to extract the numerical data, OpenCV and Google MediaPipe. Although the tracking is usually accurate on these programs, some of the videos selected were unable to be tracked accurately.
- Examples of this are shown on the following page

Discussion (cont.)

-  
- The first two images show example of the tracking program running correctly, and the third image shows some of its limitations. When the subject is well-framed against a light or neutral background, OpenCV and MediaPipe have no issues picking up the nuances of their body movements. One important thing to note is that the relationship between the heel and the toe, which is a critical aspect of a runner's form but a nuanced detail for the CV programs to catch, is reflected accurately in the first two images. This is essential because it was a feature that was considered during the classification of the videos into good or bad categories.
- The third image shows the occasional unreliability of these programs, which could skew results depending on whether good or bad videos were incorrectly extracted by the CV programs. In this case, both good and bad videos were incorrectly extracted throughout the course of the extraction process, which explains the mediocre test accuracy, but the precision and recall being relatively close in value.
- One idea moving forward is to process the videos before they are fed to OpenCV and MediaPipe, which could include increasing the contrast, or using some sort of convolutional layer to highlight any moving object. This would make the figure in the video pop out more, even in poorly lit or awkwardly framed conditions, which would improve the accuracy of the numerical extraction by the CV programs.

Conclusions

- At this point the research goal has not been met, which was 85% test accuracy for this model. Currently the test accuracy rests at 67%. However, the Discussions section points out the numerous ways in which this value could be increased, which would turn the technology into a viable and novel new opportunity for gait analysis without a kinematic sensor, which is common in other gait analysis projects which include machine learning (Zhang et. al, 2020).
- The primary application for this work would be gait analysis for amateur runners seeking to improve their form in an accessible and affordable way. However, it could also have application in research contexts where an alternative to using kinematic sensors is desired, as this approach to gait analysis leaves the runner unrestricted during their run.

Part 7 - References

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